



Department of Computer Science and Engineering

Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8602 Compiler Design

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic:** Unit II / Context free grammars, Writing a Grammar

- **Course Outcome:** CO2

- **Topic Learning Outcome:** TLO4

- **Activity Chosen:** Flipped Classroom

- **Justification:**

The basics of context free grammar and how to write a grammar is very much needed for understanding the functionalities of syntax analyzer and performing parsing. They have already studied in theory of computation subject in previous semester. Hence to recollect and get a sound knowledge in the topic by viewing the material and have elaborate discussion in class room, this flipped classroom activity is planned.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

- The learning materials were posted to the students in canvas two days before the topic discussion.
- The students are instructed to study the material and take notes on the topic based on their understanding.
- On the day of the activity the students were grouped into a team of 4 members and they are asked to discuss about the topic they learnt in their home through the study materials for the duration of 10 minutes and it is shown in Fig.1.
- Based on the study material they have gone through, the problems were given for each group separately and by discussing with their team members they have solved it.
- After the discussion one member from each team had a chance to present and share the way of solving the problems to the entire class as shown in Fig.2 & 3.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	2	2	2	1	2	1	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1	PSO3
	3	2	2	2	2	2	1
Justification	The students	The	The	The students	The	The students	The

for correlation	apply the knowledge of context free grammar to define grammar for a particular programming structure	students analyze the given questions and identify solutions	students find the solutions for the given question.	effectively communicate their answers	students are working as a team to provide the conclusion	use the context free grammar concepts for defining the programming syntax	students use the knowledge of grammar to define AI based compiler
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• **Images / Screenshot of the practice:**



Fig.1. Students discuss with their team members



Fig.2. Solving the problem



Fig1. Student share their view with classmates

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Students realized the benefits of self-learning while discussing with their group members.
 - Those who didn't get clear ideas from the study material, with the help of peer team learning they understood the concepts clearly.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- The students get more clarity in the topic of context free grammar due to their self-learning and knowledge sharing with their peer group.
 - Most of the students (47 members) attended and scored marks in the Part A- Question No.4, which is based on context free grammar derive the string, in Internal Assessment Test- I.

- ❖ **Challenges faced in implementation:**

- It is difficult to make all the students to go through the study material hence they struggle a lot to recollect the topic and not able to interact effectively among the team members.
 - Due to the small size of the team, few teams cannot get a chance to share their views.

References:

- ❖ <https://www.ritrjpm.ac.in/images/computer-science/Flipped-Classroom-PJ-DAA.pdf>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/CS8603-FlippedClassroom.pdf>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/32_CS6703_Flippedclassroom.pdf

Signature of Faculty Member

HOD